Algebra Review: Unit 6 — Exponents & Exponential Functions

Station 1 of 4: Properties of Exponents

Name: _____ Date: _____

Simplify each expression. Write answers using positive exponents.

1. Simplify: $x^3 \cdot x^4$

2. Simplify: $(2a^3)(5a^2)$

3. Simplify: $\dfrac{x^9}{x^4}$

4. Simplify: $(m^2)^5$

5. Simplify: $(3y^2)^3$

6. Simplify: $\dfrac{12a^5b^3}{4a^2b}$

7. Simplify: $(2x^3y^2)^2 \cdot x$

8. Simplify: $\left(\dfrac{2x}{y^3}\right)^4$

9. Simplify: $(-3x^2)^3$

10. Simplify: $\dfrac{(x^3)^4}{x^5}$

Algebra Review: Unit 6 — Exponents & Exponential Functions

Station 2 of 4: Zero and Negative Exponents; Scientific Notation

Name: _____ Date: _____

Evaluate or simplify. Write final answers with positive exponents. Use scientific notation where asked.

1. Evaluate: $5^0 + 7^0 + (-8)^0$

2. Evaluate: 2^{-3}

3. Evaluate: $\left(\frac{1}{3}\right)^{-2}$

4. Simplify: x^{-5}

5. Simplify: $\frac{3a^{-2}}{b^{-4}}$

6. Simplify: $(2x^{-3})^{-2}$

7. Write 4,500,000 in scientific notation.

8. Write 0.00032 in scientific notation.

9. Multiply and write the answer in scientific notation: $(4 \times 10^3)(5 \times 10^6)$

10. Divide and write the answer in scientific notation: $\frac{8 \times 10^7}{2 \times 10^3}$

Algebra Review: Unit 6 — Exponents & Exponential Functions

Station 3 of 4: Exponential Functions: Evaluating and Identifying

Name: _____ Date: _____

Answer each question about the exponential function or table. Diagrams are not drawn to scale.

1. Evaluate $f(x) = 3 \cdot 2^x$ at $x = 4$.

2. Evaluate $f(x) = 5\left(\frac{1}{2}\right)^x$ at $x = 3$.

3. Evaluate $f(x) = 2^x$ at $x = -2$.

4. Is $y = 4(1.5)^x$ exponential growth or decay? Name the growth factor.

5. Is $y = 200(0.8)^x$ exponential growth or decay? Name the decay factor.

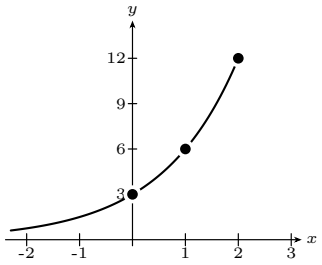
6. State the y-intercept and the growth factor of $y = 10(3)^x$.

7. Complete the table for $y = 3^x$: $x = 0, 1, 2, 3$.

8. Does the table show an exponential function? x: $0, 1, 2, 3$ y: $2, 6, 18, 54$. If so, write the equation.

9. Does the table show an exponential function? x: $0, 1, 2, 3$ y: $4, 8, 12, 16$.

10. Write an exponential function for the graph shown. It passes through the marked points.



Algebra Review: Unit 6 — Exponents & Exponential Functions

Station 4 of 4: Exponential Growth and Decay Problems

Name: _____ Date: _____

Use $y = a(1 \pm r)^t$. Round money to the nearest cent and other answers to the nearest whole number unless told otherwise.

1. A population of 500 grows 4% per year. Write a function and find the population after 3 years.

2. A car worth \$20,000 loses 15% of its value each year. What is it worth after 4 years?

3. Find the balance after 10 years if \$1,000 is invested at 5% interest compounded annually.

4. A bacteria culture of 100 doubles every hour. How many bacteria are there after 6 hours?

5. A 80 g sample of a substance has a half-life of 8 days. How much remains after 24 days?

6. A town of 12,000 people declines 2% per year. Write a function for the population P after t years.

7. Use your function from the previous problem to estimate the population after 5 years.

8. For each function, state the percent growth or decay per period: (a) $y = 250(1.07)^t$ (b) $y = 90(0.92)^t$

9. Find the balance after 2 years if \$3,000 is invested at 4% compounded quarterly: $A = P\left(1 + \frac{r}{n}\right)^{nt}$.

10. A collectible worth \$2,000 doubles in value every 5 years. What is it worth after 15 years?

Algebra Review: Unit 6 — Exponents & Exponential Functions

Answer Key

Station 1: Properties of Exponents

1. x^7
2. $10a^5$
3. x^5
4. m^{10}
5. $27y^6$
6. $3a^3b^2$
7. $4x^7y^4$
8. $\dfrac{16x^4}{y^{12}}$
9. $-27x^6$
10. x^7

Station 2: Zero and Negative Exponents; Scientific Notation

1. 3
2. $\dfrac{1}{8}$
3. 9
4. $\dfrac{1}{x^5}$
5. $\dfrac{3b^4}{a^2}$
6. $\dfrac{x^6}{4}$
7. 4.5×10^6
8. 3.2×10^{-4}
9. 2×10^{10} $\quad 20 \times 10^9$
10. 4×10^4

Station 3: Exponential Functions: Evaluating and Identifying

1. 48
2. $\dfrac{5}{8}$
3. $\dfrac{1}{4}$
4. Growth; growth factor 1.5 *since $b > 1$*
5. Decay; decay factor 0.8 *since $0 < b < 1$*
6. y-intercept $(0, 10)$; growth factor 3
7. $y = 1,\ 3,\ 9,\ 27$
8. Yes (constant ratio 3); $y = 2 \cdot 3^x$
9. No: constant differences (linear, $y = 4x + 4$)
10. $y = 3 \cdot 2^x$ $\quad a = 3$*; growth factor* $\dfrac{6}{3} = 2$

Station 4: Exponential Growth and Decay Problems

1. $P = 500(1.04)^t;\ P(3) \approx 562$
2. $20{,}000(0.85)^4 \approx \$10{,}440.13$
3. $1000(1.05)^{10} \approx \$1{,}628.89$
4. $100 \cdot 2^6 = 6400$
5. $80\left(\dfrac{1}{2}\right)^3 = 10$ g $\quad \dfrac{24}{8} = 3$ *half-lives*
6. $P = 12{,}000(0.98)^t$
7. $12{,}000(0.98)^5 \approx 10{,}847$
8. (a) 7% growth (b) 8% decay
9. $3000\left(1 + \dfrac{0.04}{4}\right)^8 \approx \$3{,}248.57$
10. $2000 \cdot 2^3 = \$16{,}000$